

AV_Culpability_E18_Behavior_MultiTasking_Notifs - SHARED

Start of Block: informed_consent

ic Informed Consent Participation is voluntary It is your choice whether or not to participate in this research. If you choose to participate, you may change your mind and leave the study at any time. Refusal to participate or stopping your participation will involve no penalty or loss of benefits to which you are otherwise entitled. **What is the purpose of this research?** The purpose of this research is to examine human judgments about automated vehicles. All data from this experiment are gathered for scientific purposes and will contribute to our eventual understanding of brain and visual function. These data may be published in scientific journals so that other researchers may have access to these data. **How long will I take part in this research?** Your participation will take approximately 7 minutes to complete. **What can I expect if I take part in this research?** As a participant, you will be asked to look at images presented on a video display and give responses with key presses or movements of a mouse pointing device. Your response may involve responding as quickly as you can, memorizing what you saw, making a judgment, or completing a questionnaire. You will also be asked to complete a demographics form. **What are the risks and possible discomforts?** If you choose to participate, the effects should be comparable to those you would experience from viewing a computer monitor for 7 minutes and using a mouse or keyboard, e.g., eye fatigue. You are free to take breaks throughout the session. **Will I be compensated for participating in this research?** You will be compensated \$0.94 for this study. You will still receive payment if you withdraw early. **If I take part in this research, how will my privacy be protected?** **What happens to the information you collect?** Your participation in this experiment will remain confidential, and your identity will not be stored with your data. **If I have any questions, concerns or complaints about this research study, who can I talk to?** The researcher for this study is Julian De Freitas who can be reached at 626.559.6401; #161 Morgan Hall, 15 Harvard Way, Boston MA, 02163; jdefreitas@hbs.edu. If you have questions, concerns, or complaints, If you would like to talk to the research team, If you think the research has harmed you, or If you wish to withdraw from the study. This research has been reviewed by the Committee on the Use of Human Subjects in Research at Harvard University. They can be reached at 617-496-2847, 1350 Massachusetts Avenue, Suite 935, Cambridge, MA 02138, or cuhs@harvard.edu for any of the following: If your questions, concerns, or complaints are not being answered by the research team, If you cannot reach the research team, If you want to talk to someone besides the research team, or If you have questions about your rights as a research participant.

☐ I agree (1)

☐ I disagree (2)

End of Block: informed_consent

Start of Block: Prolific ID



prolific_id What is your Prolific ID? Please note that this response should auto-fill with the correct ID.

End of Block: Prolific ID

Start of Block: attention_checks

att_1 **What color is grass? The fresh, uncut grass, not leaves or hay.** Please make sure to select purple, so that we know you're paying attention.

☐ Green (1)

☐ Purple (2)

captcha Click on the checkbox below

End of Block: attention_checks

Start of Block: attention check fail

Q143 Unfortunately, you have failed to answer at least one of the attention check questions correctly and have been redirected to the end of the survey. Please click the button to below to be redirected back to Prolific and register your submission.

End of Block: attention check fail

Start of Block: av_general

av_info **Please read the information presented below carefully in order to be able to understand and answer the survey questions Below is a table depicting 6 possible types of vehicles. The different types are based on how many driving tasks are controlled by the human driver versus the self-driving system.** In level 1, the human controls all driving tasks, whereas in level 6, the self-driving system controls all driving tasks. The levels in-between differ in which driving tasks are controlled by the human versus the self-driving system.

comp_1 **You were shown a table. What did the table represent?**

- ☐ The number of typical daily tasks controlled by humans vs. machines (1)
 - ☐ Different types of vehicles based on how many driving tasks are controlled by humans vs. the self-driving system (2)
 - ☐ Six people's preferences for humans vs. machines (3)
-

comp_2 **What were the values for vehicle type #1 (the one at the top of the table)?**

- ☐ SYSTEM, HUMAN, HUMAN, SYSTEM (1)
 - ☐ HUMAN, SYSTEM, SYSTEM, SYSTEM (2)
 - ☐ SYSTEM, SYSTEM, SYSTEM, SYSTEM (3)
 - ☐ HUMAN, HUMAN, HUMAN, HUMAN (4)
-

Page Break

aeon_info AEON is an automobile company that is planning on introducing two new automated vehicle programs in the market. On the next two pages we will ask you a question about each program.

End of Block: av_general

Start of Block: av_frame1

frame1_info1 One of Aeon's automated vehicle programs is called: **AEON AUTOPILOT**. Please answer some questions so that AEON can understand and analyze preferences for automated vehicles and investigate the potential of AEON AUTOPILOT.

Page Break

auto_1 What do you perceive as the level of automation for AEON's AUTOPILOT Program?

☐ 1 (1)

☐ 2 (2)

☐ 3 (3)

☐ 4 (4)

☐ 5 (5)

☐ 6 (6)

frame1_info2

Page Break

End of Block: av_frame1

Start of Block: av_frame2

frame2_info1 One of Aeon's automated vehicle programs is called: **AEON COPILOT**. Please answer some questions so that AEON can understand and analyze preferences for automated vehicles and investigate the potential of AEON COPILOT.

Page Break

co_1 What do you perceive as the level of automation for AEON's COPILOT Program?

☐ 1 (1)

☐ 2 (2)

☐ 3 (3)

☐ 4 (4)

☐ 5 (5)

☐ 6 (6)

frame2_info2

End of Block: av_frame2

Start of Block: task_instructions_notif

instructions

You will now watch a simulation in which the vehicle will drive using one of AEON's two programs--depending on your choice.

The vehicle's performance will directly determine your **bonus payment**. If the vehicle hits **less than 5 blocks**, you will be rewarded with a bonus of **\$0.30**. At the same time, you will be receiving a series of **smartphone notifications**. Each time you receive one, they cover most of your view (as depicted below). Your sole task is to decide whether you want the vehicle to drive using **AUTOPILOT or COPILOT**. Based on your choice, the vehicle will drive with the chosen system. During this time, you **do not need to interact** with the simulation. Now, please answer the following comprehension check, then make your selection:



comp_notif What two factors cause you to lose points in the simulation?

- ☐ Using autonomous driving & time elapsed (1)
- ☐ Time elapsed & hitting white obstacles (2)
- ☐ Running out of fuel & driving too quickly (3)



adas_choice_notif **Which driving system would you like to choose for the simulation?**

- ☐ **AEON Copilot** (1)
- ☐ **AEON Autopilot** (2)

End of Block: task_instructions_notif

Start of Block: simulation_auto_notif



sim

Please watch the simulation below. Go to the next page after you're done.

End of Block: simulation_auto_notif

Start of Block: simulation_co_notif



sim

Please watch the simulation below. Go to the next page after you're done.

End of Block: simulation_co_notif

Start of Block: task_instructions_plain

instructions

You will now watch a simulation in which the vehicle will drive using one of AEON's two

programs--depending on your choice.

The vehicle's performance will directly determine your **bonus payment**. If the vehicle hits **less than 5 blocks**, you will be rewarded with a bonus of **\$0.30**. Your sole task is to decide whether you want the vehicle to drive using **AUTOPILOT or COPILOT**. Based on your choice, the vehicle will drive with the chosen system. During this time, you **do not need to interact** with the simulation. Now, please answer the following comprehension check, then make your selection:



comp_plain What two factors cause you to lose points in the simulation?

- ☐ Using autonomous driving & time elapsed (1)
- ☐ Time elapsed & hitting white obstacles (2)
- ☐ Running out of fuel & driving too quickly (3)



adas_choice_plain Which driving system would you like to choose for the simulation?

- ☐ AEON Copilot (1)
- ☐ AEON Autopilot (2)

End of Block: task_instructions_plain

Start of Block: simulation_auto_plain



sim

Please watch the simulation below. Go to the next page after you're done.

End of Block: simulation_auto_plain

Start of Block: simulation_co_plain



sim

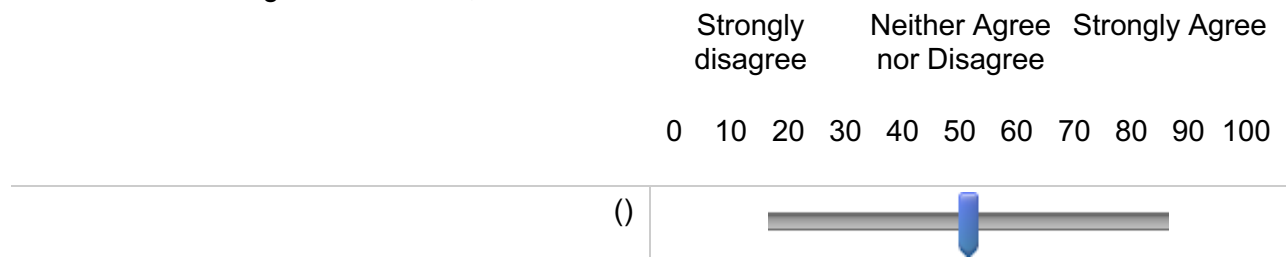
Please watch the simulation below. Go to the next page after you're done.

End of Block: simulation_co_plain

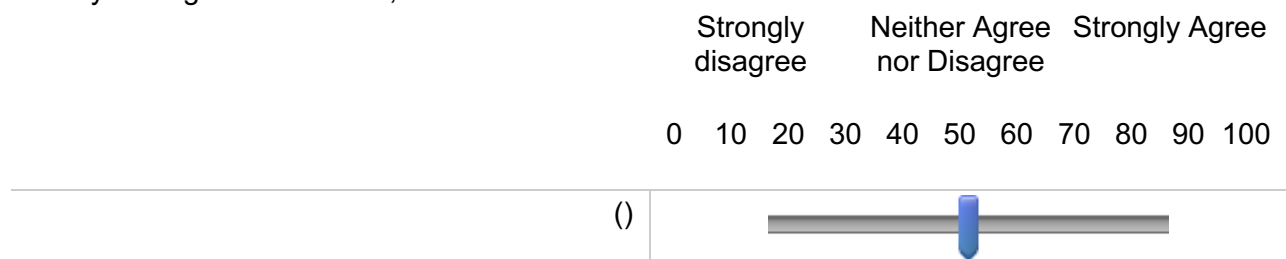
Start of Block: comfort

comfort_instructions Thinking about the simulation you just watched, please indicate how much you agree with the following statements

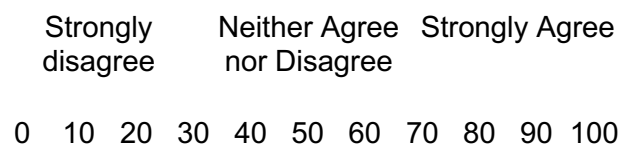
relaxedness During the simulation, I felt relaxed



safety During the simulation, I felt safe



anxiety During the simulation, I felt anxious



()



End of Block: comfort

Start of Block: demographics

gender What is your gender?

- ☐ Male (1)
 - ☐ Female (2)
 - ☐ Prefer not to disclose (3)
 - ☐ Other (please specify) (4)
-

ethnicity What is your ethnicity?

- ☐ Black or African American (1)
 - ☐ Asian (2)
 - ☐ White or European (3)
 - ☐ Hispanic or Latino (4)
 - ☐ Mixed (5)
 - ☐ Prefer Not to Say (7)
 - ☐ Other (please specify) (6)
-

age What is your age (in years)?

education Please indicate the highest level of education completed.

- ☐ High School or Equivalent (1)
- ☐ Vocational/Technical School (2 year) (2)
- ☐ Some College (3)
- ☐ College Graduate (4 year) (4)
- ☐ Masters Degree (MS) (5)
- ☐ Doctoral Degree (PhD) (6)
- ☐ Professional Degree (MD, JD, etc.) (7)
- ☐ Other (please specify) (8)

license **Do you have a driving license?**

- ☐ Yes (1)
- ☐ No (2)

ai_knowledge **How much would you say you know about automated vehicles?**

Very little

A lot

0 10 20 30 40 50 60 70 80 90 100

()



End of Block: demographics
